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AN2DL - Second Challenge Report
Marcus.ai

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1 Introduction

In this paper, we address molecular subtype prediction from histopathology imagery. Whole-slide histopathology images (WSIs) are large, heterogeneous, and affected by staining and acquisition variability, which makes direct slide-level CNN learning difficult [2]. Therefore, many WSI pipelines use patch-based learning and attention-based learning to focus on informative regions [1, 3].

In our task, each input is a low-magnification patch extracted from a WSI, and the goal is to classify it into **Luminal A**, **Luminal B**, **HER2(+)**, or **Triple Negative**. An optional binary mask highlights tissue regions of interest and can reduce background influence. We formulate the problem as **supervised multi-class classification** and evaluate performance using the **F1-score**.

Our focus is robustness to stain variability, tissue heterogeneity, patch-scale changes, and especially dirty images with noisy background and artifacts, to improve generalization to held-out test data.

2 Problem Analysis

This task is a four-class **supervised multi-class classification** problem on low-magnification WSI patches. The main evaluation metric is the **F1-score**, so the model must balance precision and recall, not only overall accuracy.

A key difficulty is that discriminative morphology can be subtle and localized, while large parts of a patch may be weakly informative. In addition, the dataset contains very **dirty images with noisy background** and artifacts, which can push the model to learn spurious correlations and reduce generalization. This is consistent with known WSI deployment challenges: published validation studies report technical issues such as scanning failures, prolonged turn-around times, and the need for higher magnification, as well as diagnostic difficulties for fine-grained tasks [4]. In our setting, low magnification and variable image quality can similarly limit visibility of fine patterns and make the model sensitive to acquisition and preparation artifacts.

An **binary mask** is provided as auxiliary information to highlight tissue regions of interest. When used, it can reduce background influence and improve robustness. Therefore, the main focus is to build a model that generalizes well under noise, stain variability, and tissue heterogeneity, while optimizing **F1-score** on held-out data.

3 Method

3.1 Pre-processing

We design a pre-processing pipeline to remove junk data, reduce artifacts, and create stable inputs for training.

First, we remove **outlier (junk) images** using HSV color analysis by detecting abnormal yellow–green–cyan artifact pixels. Images with a high artifact ratio are flagged; after manual inspection, flagged images (and their masks) are removed and labels are updated.

Second, we remove strong color artifacts by detecting green stain and bright/shiny pixels and replacing them with a gray background plus small noise to avoid misleading patterns.

Third, we use the provided **binary mask** to focus on tissue. We apply **Mask** $\times$ **Image** to keep pixels inside the ROI. Pixels outside the ROI are not kept as black; instead, we fill them with the same gray value $(186, 186, 186)$ plus noise (**Mask** $\times$ **Image + Gray BG**). This makes the background closer to typical slide background and reduces spurious correlations.

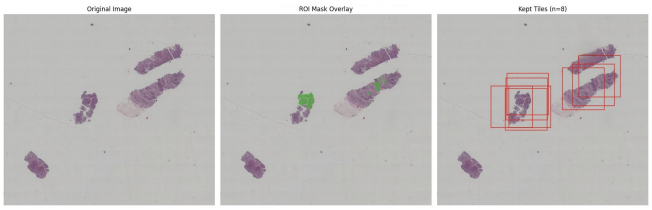


Figure 1: Pre-processing overview: original patch, ROI mask overlay, and the selected tissue-rich tiles.

Fourth, we perform **ROI-guided tiling**. Using the mask, we compute the ROI bounding box and generate candidate tiles around the ROI (center/corners) and on a coarse grid. We then select a small set of tiles per image using a greedy strategy that prioritizes high ROI coverage, so the final tiles contain as much informative tissue as possible.

Finally, we apply **tile quality filtering**. For each tile, we estimate how much of the tile belongs to the ROI and how informative the visual content is using a simple variance check. Tiles with very little tissue or very low information content are discarded, so the model trains on cleaner and more meaningful regions.

3.2 Models

Input to the model. We use 224×224 RGB tiles extracted from tissue ROI after pre-processing. Inputs are normalized with ImageNet mean/std. For Model 2, we also provide the aligned **binary mask** as an extra input channel, so the network can use ROI geometry and reduce background influence.

Loss function. We train with multi-class CrossEntropyLoss. To handle possible class imbalance, we apply class weights computed from the training label distribution. (Model 2 also supports label smoothing.)

Optimization and fine-tuning. Model 1 uses AdamW with staged fine-tuning (train the head with a frozen backbone, then fine-tune the last block). Model 2 uses **Ranger**; the mask branch and fusion head are trained from the start, while the RGB backbone is fine-tuned more conservatively.

Data augmentation. We use lightweight spatial augmentation (random flips and small affine transforms). Augmentation is applied to the RGB tile, and the mask remains binary and aligned.

Validation and inference aggregation. Training is tile-level, but evaluation is image-level. We average class probabilities over all tiles of an image and predict by argmax; the same aggregation is used at test time with simple TTA.

3.2.1 Patch-based ResNet classifier

Network architecture. Model 1 uses a pretrained ResNet backbone with a small classification head for 4 classes. We test two variants:

- **ResNet50** $\rightarrow$ Global Average Pooling $\rightarrow$ Linear$(d, 4)$ with **Dropout**$(p = 0.1)$ in the head $\rightarrow$ Linear$(d, 4)$
- **ResNet101** $\rightarrow$ Global Average Pooling $\rightarrow$ Linear$(d, 4)$ with **Dropout**$(p = 0.4)$ in the head $\rightarrow$ Linear$(d, 4)$

3.2.2 Model 2: Dual-Path EfficientNet-B2 with masks

Network architecture. Model 2 is a two-branch (**Dual-Path**) architecture that uses both the RGB tile and the binary mask:

- **RGB branch:** a pretrained **EfficientNet-B2** encoder extracts a compact feature vector from the RGB tile.

- **Mask branch:** a lightweight CNN encodes the binary mask to capture ROI geometry and tissue layout.

The two feature vectors are fused by concatenation in latent space, followed by a small MLP classifier (**Dropout** → Linear → ReLU → **Dropout** → Linear) to produce 4-class logits. The main motivation is to let the model combine visual texture (RGB) with spatial ROI information (mask), especially under noisy background and artifact conditions.

4 Experiments

We evaluate on the WSI patch dataset with four molecular subtypes. All models are trained on ROI-derived tiles from our pre-processing and tiling pipeline, and we report macro **F1-score**.

Baselines. We first tried a **non-patching** classifier and several alternative ROI/pre-processing ideas including a **U-Net**-style segmentation pipeline. We also tested multiple cleaning/normalization variants such as, **Morphological Masking**, and **Macenko Normalization**, but these approaches did not pass **36%** test F1. This motivated a patch-based setup where discriminative morphology is better captured from tissue-rich regions.

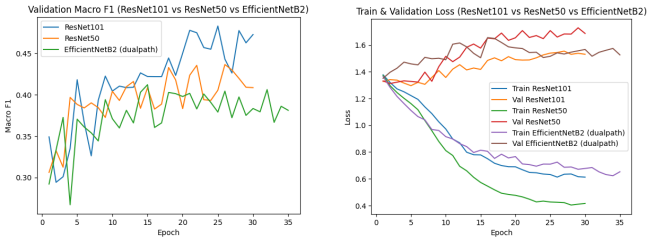
Final models. Our final experiments compare patch-based **ResNet50/ResNet101** and a **Dual-Path EfficientNet-B2** that uses both RGB tiles and the binary mask.

Table 1: Validation and test F1-score for the evaluated models. Best test result is in **bold**.

Model	Val F1 (%)	Test F1 (%)
EfficientNet-B2 (Dual-Path)	41.20	**40.65**
ResNet50	43.64	40.26
ResNet101	48.29	39.47
ResNet50 (without patching)	39.37	35.85

5 Results

Figure 2 compares validation macro F1 and train/validation loss. ResNet101 reaches the highest validation F1, but the Dual-Path model achieves the best test F1, suggesting better robustness to real test noise.



(a) Validation macro F1 (b) Train/validation loss

Figure 2: Model comparison on the validation set.

Note. Training loss is computed on patches, while validation loss is computed on images after tile aggregation.

6 Discussion

Strengths. The ROI-focused patch pipeline reduces background bias and helps learning from tissue-rich regions in dirty images. The Dual-Path EfficientNet-B2 benefits from using the **binary mask** as auxiliary input, improving robustness when RGB appearance is noisy.

Weaknesses. Generalization remains difficult due to stain variability, artifacts, and heterogeneous tissue content. Simple probability averaging for aggregation is stable but may underweight highly informative tiles.

7 Conclusions

Non-patching baselines and alternative ROI/pre-processing strategies stayed below 36% test F1. A patch-based pipeline with artifact-aware pre-processing improved performance, and the **Dual-Path EfficientNet-B2** achieved the best test F1 by combining RGB texture with mask-based ROI information.

References

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